Time: 1 hour B.Tech V Semester, Sessional Test 2, 2023-24 Max Marks: 15

Object Oriented Programming Systems (OOPS), CEN - 504

Q1 (1+1+3) 5 Marks Attempt ALL questions. CO3

(a) *Inheritance enables run-time polymorphism.* Justify this statement.

(b) Define a class **Element** with no data members and two member functions, **void print()** and **double norm()**, a default constructor and a default destructor. **print** should be virtual while **norm** should be pure virtual. Can you instantiate an object of class **Element**? Why or why not?

(c) Define a class **PointND** with a single data member which is a pointer to **double**. **PointND** is derived publicly from **Element**. Provide a default constructor, a parameterized constructor that accepts a **double** array and its size **aSize** (type **int**) and initializes the object with the data contained in the array, and a default destructor. Provide the class definition and implementation.

Q2, (2.5+2.5) 5 Marks CO3, CO4

(a) Will the following code compile and run without problems in C++? What if similar code was written in Java? Explain.

```
double someData[3] = {1,2,3};
Element * anElementPtr = new PointND(someData,3);
delete anElementPtr;
```

(b) Let's say you have a class hierarchy in Java where the abstract base class is **Vehicle** with an abstract function **print** and two derived classes **Bicycle** and **Car**. You are given a reference myVehicle. **Bicycle** has a function "**pedal**" and **Car** has a function "**drive**". Define and implement this hierarchy.

Q3. 5 Marks CO4

Let's say you have an **ArrayList** that can hold objects of type **Object**. A vehicle, **oneVehicle**, can be added to the list by "**listOfShapes.add(oneVehicle);**". The number of vehicles is obtained by "**listOfShapes.size()**". To get the i-th object from the list call "**listOfShapes.get(i)**" (i starts at 0). This method returns an **Object**. Write a function that traverses a pre-existing list of **m** vehicles and pedals if the vehicle is a **Bicycle** and drives if **Car**.